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# Using pseudo-panels to analyse labour market transitions

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# Abstract

This study describes and discusses methods that can be used to analyse labour market dynamics in the absence of true panel data. The so-called pseudo-panel models are particularly useful to investigate how the working conditions of different demographic groups change over time or due to aggregate shocks and policies. Three standard approaches are briefly described: the age-period-cohort model, to estimate lifecycle patterns and long-term trends of labour market indicators; the mobility model based on cohort means, to examine how the gap between demographic groups varies over time; and the differences-in-differences, to estimate the sensitivity of demographic groups to macro shocks and policies. Still, neither of these models can fully assess individual transitions. Some components of individual transitions, such as unemployment risk, income volatility, and within-cohort mobility, can only be measured using true panels. However, to examine how much individual mobility is potentially hidden in a pseudo-panel analysis, this paper also discusses the application of the synthetic panel approach, along with an exercise using data from Brazil. Finally, the paper presents methods to construct optimal-size cohorts and improve the accuracy of synthetic panels.

Keywords: Labour market dynamics, pseudo-panel, synthetic panel, cohort analysis, cohort construction

JEL: C23, C81, J21, J69

# 1. Introduction[1]

The assessment of labour mobility is as critical as labour market figures for the design of public policies. To improve labour conditions, policymakers must understand whether at the individual level, these conditions are permanent or just transitory. For instance, whether youth unemployment hinder future opportunities or not implies distinct ways to handle the issue. Similarly, when the overall conditions improve, policymakers must anticipate who are the most vulnerable to potential downturns.

A comprehensive assessment of labour mobility requires the use of panel data, which follow the same workers over time. This sort of data, though, is not widely available, especially in developing countries. To overcome the lack of panel data, many studies since Deaton (1985) have proposed pseudo-panel approaches using multiple rounds of cross-sectional surveys.

In this paper, I summarize four approaches that can be applied for the investigation of labour mobility using repeated cross-sections (RCS): the **age-period-cohort** (APC) model, useful for the estimation of lifecycle and long-term trends of static rates, such as unemployment, college attainment, and informality; the dynamic model of **cohort means**, which examines how the gap between demographic groups varies over time; the **synthetic panel**, which reveals how much individual mobility is potentially hidden by the cohort-mean approach; and the **differences-in-differences** (DiD), which estimates the sensitivity of demographic groups to aggregate shocks and policies.

The uses and limitations of these four approaches are described in Section 0. However, before that, Section 0 introduce a few basic concepts applied to labour market dynamics, and Section 0 presents those that can be measured using pseudo-panels. Section 0 describes how to construct cohorts (or homogeneous groups) given the sample size and presents an exercise comparing a true panel and a pseudo-panel. Section 0 concludes the paper.

# 2. Basic Concepts

To investigate labour market transitions, we must consider that there are three general forms of transition: **raw transition**, **risk**, and **mobility**. A ***raw transition*** happens when we simply observe the same individual in two different states, from one year to another. For example, a transition rate in labour markets can be the proportion of individuals who were employed when first observed and unemployed one year later.

However, the raw transition rate depends on two events. The first one regards the fact that, in the same year, a worker can change its status many times. For instance, consider two workers, Josh and Robert, observed in the same year. Given his skills and seasonal economic activities, Josh is often formally employed in the first quarter, but he is self-employed for the rest of the year. Robert is formally employed in the third quarter and self-employed in the other quarters. Hence, these two workers have similar working status, but the assessment of their conditions depends on when they are interviewed. These changes within the same period are defined as ***risk***, which can also result from measurement errors. If workers are interviewed only once a year, the researcher cannot double check whether the information reported is consistent. If this information is misreported or somehow volatile, it tends to inflate the raw transition rate.

The second event is ***mobility***, which is defined as the change in a worker's expected status. For example, consider a worker who spends six months in the formal sector and six months in the informal sector. In the following year, if the worker spends more than six months in the informal sector, they experience mobility into the informal sector. If the purpose is to investigate long-run transitions, the mobility rate is a more appropriate measure than the raw transition rate. These two measures will be similar only if the risk (or measurement error) is zero, which is not often the case when analysing vulnerable workers. Accordingly, we focus on mobility as a measure of interest hereafter.

[1] This background paper aims at discussing the methodology of pseudo-panels to analyse labour market transitions. It will be complemented by an empirical study afterwards.

# 3. What Can be Measured with a Pseudo-Panel?

A mobility rate can be decomposed into **net cohort mobility** and **within-cohort mobility**. A ***cohort*** is defined as a group of individuals sharing the same attributes such as date of birth, biological sex, and socioeconomic background. If these attributes do not vary over time, then changes in the (static) behaviour of cohort members can be assessed by the difference in cohort means between periods (Deaton, 1985).

The ***net cohort mobility*** is the increase in the share of cohort members in a certain state between two periods. This increase must be balanced by a reduction in the share of members in other states. For example, if the labour force participation of a cohort increases from 20% to 40%, the share of members out of the labour force must decrease from 80% to 60%. The resulting net mobility of 20 percentage points (p.p.) means that at least 20% of this cohort has moved into the labour force.

Still, could more than 20% move into the labour market? Yes, as long as they switch their status with other members of the same cohort. For example, suppose that a total of 30% of the cohort moves into the labour force. Then, while 20% have switched with members of other cohorts (i.e., the net mobility rate), 10% have switched with members of the same cohort. This residual rate is defined as the ***within-cohort mobility***. That is, the share of cohort members in a certain state who are replaced by members in another state, without changing the overall share of members in each state.

As discussed in Section 0, pseudo-panels provide reliable estimates for the net cohort mobility, which is useful for the analysis of demographic groups. However, they are very limited in the estimation of within-cohort mobility.

# 4. The Use of Pseudo-Panels to Assess Mobility

When a true panel data set is not available, one may estimate panel data models and statistics using RCS. In this approach, individuals are grouped into cohorts that are treated as observations in a ***pseudo-panel*** (Deaton, 1985; Verbeek, 1996). On the one hand, a pseudo-panel does not suffer from the attrition problem, since new samples are drawn every period (Deaton, 1985). Also, it yields mobility estimates that are consistent under measurement errors (Antman & McKenzie, 2007). On the other hand, as variables are aggregated by cohorts and periods, the identification of individual relationships requires very particular conditions (Robinson, 1950; Verbeek, 2008). Due to the embedded ecological fallacy, pseudo-panels tend to underestimate mobility rates. In this section, we explore four pseudo-panel approaches and discuss their assumptions.

The models presented in this section are applied for different purposes. See Table 1 for a summary.

**Table 1 – Summary of Pseudo-Panel Models**

| Model | Purpose | Example | Required data |
|---|---|---|---|
| Age-Period-Cohort | To estimate lifecycle and long-term trends of static rates | Female labour force participation rates | Long series of RCS |
| Cohort-Mean | To verify how definite or malleable the gap between demographic groups is | Net cohort income mobility and poverty dynamics | Large number of cohorts with enough observations |
| Synthetic Panel | To estimate bounds for individual mobility (or how much mobility is potentially hidden in the net cohort rate) | Lower and upper bounds for poverty dynamics | Large number of cohorts with enough observations |
| Difference-in-Difference | To estimate the sensitivity of cohorts to aggregate shocks | The heterogenous impact of the COVID-19 pandemic across demographic groups | RCSs with enough frequency around the event |

## 4.1. The Age-Period-Cohort Model

For the purpose of estimating the **lifecycle and long-term trends of static rates** (e.g., unemployment, college attainment, poverty, and informality), the APC model is the most appropriate. Because it is the only

model that fully accommodates all the events that determine the difference between younger and older individuals (Mason, et al., 1973). Still, the APC model is not directly used to estimate transitions and mobility. Instead, it should be viewed as a benchmark to analyse cohort rates.

Formally, the APC model can be written as follows:

$$y_{igt} = \alpha_g(age_{igt}) + \beta(period_{ig}) + \gamma(cohort_{ig}) + u_{igt}, \quad (1)$$

where $y_{igt}$ is the outcome of interest, which can be either continuous or binary, and $u_{igt}$ is a mean-zero, random term of individual $i$ from birth cohort $g$ in period $t$. This equation also has three additive functions, which can be estimated linearly, non-linearly, or discontinuously (with dummies).

The first function, $\alpha_g(.)$, represents the effect of lifecycle events. A simpler, restricted model would assume that these effects do not vary across cohorts — i.e., for all $g$, $\alpha_g(.) = \alpha(.)$ —, whereas the unrestricted model would allow cohorts to have distinct lifecycles. This heterogeneity would also comprise the fact that macro-events distinctly impact cohorts, depending on their age. For instance, a financial crisis may distinctly affect the behaviour of young and old cohorts. If these heterogeneous effects are transient, they will reshape the age function of those cohorts. Nevertheless, the identification of heterogeneous lifecycles requires a very long series of RCS, so that a wide enough range of cohorts are observed from the beginning to the end of their lifespan. For short series, one should assume that the age function is the same for all birth cohorts.

The second function, $\beta(.)$, represents short-term macro-events that affect all cohorts similarly. In social sciences, cohort effects represent the sum of unique exposures experienced by a group of individuals (Ryder, 1965). Under this conceptualization, the third term on the right-hand side of equation (1), $\gamma(.)$, captures long-term trends. The principle is that if a macro-shock has a permanent impact on individuals, it should shift the cohort function, $\gamma(.)$. For instance, a technological shock, such as automation, should permanently affect the employment and productivity of cohorts that are exposed to it.

The APC model has an estimation issue. It regards the fact that age is equal to period minus birth cohort. The perfect collinearity among the three variables makes impossible to compute coefficients for equation (1), regardless of the effects being linear or non-linear. The simplest solution for the computational problem is to assume that at least two categories of age, cohorts, or periods have identical effects. For example, one may constrain the coefficients of the model so that the two oldest cohorts are equally distinct from others.

Although the constrained coefficient approach may solve the computational issue, the strong collinearity among the covariates still makes estimates diverge from the true model. According to Deaton and Paxson (1994), the collinearity in equation (1) implies that any trend in the data can be reinterpreted as a period trend or trends in age and cohort with opposite signs. Considering that period effects should be transient and long-term trends should be captured by cohort effects, they propose to constrain the period effects to be orthogonal to a time trend and add to zero.[2]

## 4.2. The Cohort-Mean Approach

A second pseudo-panel approach is the cohort-mean, where the characteristics of individuals are averaged by cohort and period. This approach is often applied to the analysis of mobility without a true panel (e.g., Antman & McKenzie, 2007; Ribas & Machado, 2007; Cuesta, et al., 2011), but its interpretation is limited to net cohort mobility. Still, one may use it to assess **how rigid (or malleable) the gap between demographic groups is**.

We start our discussion with a simple model of mobility. For a continuous outcome (e.g., income), mobility can be assessed by estimating the following model (Jarvis & Jenkins, 1998; Fields, et al., 2003):

$$y^*_{igt} = \rho y^*_{igt-1} + u_{igt}, \quad (2)$$

[2] To constrain the period effects, Deaton (1997, p. 126) presents the following transformation:

$$p^*_t = p_t - (t-1)p_2 + (t-2)p_1,$$

where $p_t$, $p_1$, and $p_2$ are the original dummies for periods $t$, 1, and 2, respectively, and $p^*_t$ is the new covariate indicating the period. This transformation implies that periods 1 and 2 are constrained to have the same effect and the sum of period coefficients is equal to zero.

where $y^*_{igt}$ is the outcome of interest, $y^*_{igt-1}$ is its value in the previous period, and $u_{igt}$ is the error term of individual $i$ from birth cohort $g$ in period $t$.[3] In this model, mobility is assessed by $\rho$. If $\rho$ is close to one, individual outcomes are very persistent, indicating low mobility. If $\rho$ is close to zero, individual outcomes do not depend on previous values, indicating high mobility.

A critical feature of pseudo-panels is that outcomes and states in period $t$ are observed at the individual level, but their previous values (in $t-1$, $t-2$, and so on) are not. Hence, we cannot directly estimate coefficient $\rho$ in equation (2). To overcome the missing data problem, Moffitt (1993) proposes to estimate lagged values either parametrically as a function of cohort and period effects (common to all cohort members) or simply by averaging values by cohort and period. Then, in equation (2), the lagged outcome $y^*_{igt-1}$ is replaced for the predicted (or mean) lag by cohort and period, $\hat{y}_{gt-1}$:

$$y^*_{igt} = \rho\hat{y}_{gt-1} + u_{igt}. \tag{3}$$

Antman and McKenzie (2007) also argue that the following pseudo-panel model, which averages the outcomes by cohort and period on both sides of the equation, can outperform the estimation of equation (2) with a true panel:

$$\bar{y}_{gt} = \rho\bar{y}_{gt-1} + \bar{u}_{gt}. \tag{4}$$

In a true panel, $y^*_{igt}$ is measured with an error that would bias the estimation of $\rho$. This is equivalent to the difference between transition and mobility presented in Section 4. Then, as the number of observations per cohort and period increases, the measurement error in means and the resulting bias converge to zero.

Nevertheless, the regression coefficients of equations (3) and (4) will be consistent with the one in equation (2) only in cases where cohort-period effect (e.g., the age effect in the APC model) has no direct influence on $y^*_{igt}$ (Moffitt, 1993; Verbeek, 2008). In practice, this condition would hold only if within-cohort mobility is zero, and all mobility is explained by cohort attributes. Therefore, coefficient $\rho$ in equations (3) and (4) only captures net cohort mobility.

Since pseudo-panel analyses omit within-cohort mobility, they tend to show low mobility rates. See also Antman and McKenzie (2007) for income mobility and Ribas and Machado (2007) for poverty dynamics.

## 4.3. The Synthetic Panel Approach

The synthetic panel approach is an attempt to simulate the future or past state of individuals using observed characteristics and the distribution of unobservables. The difference between this approach and the previous one is that simulated values are not necessarily common to all cohort members. Beyond the cohort mean, assumptions are made on the distribution of error terms so that simulated errors are individually assigned within cohorts. Despite the limitations imposed by pseudo-panels, one can confidently use the synthetic panel approach to check **how much mobility is potentially hidden by the net cohort mobility rate**.

As shown by Dang et al. (2014), the mobility rate must be bounded between two rates. The ***lower bound*** considers only changes that are predicted by cohort attributes. That is, it is equal to the net cohort mobility because it assumes that individuals do not switch states within cohorts. The ***upper bound*** derives from the case where the within-cohort mobility is random. That is, given the cohort mean, the current state of cohort members does not depend on their previous state.

The work of Dang et al. (2014) is ground-breaking because it does not rely on strong assumptions to find where the true mobility lies. Still, the difference between upper and lower bounds can be so large that they become uninformative. This difference is inversely related to how much individual states can be explained by cohort attributes (and directly related to how much is explained by other characteristics and events). Thus, the difference between bounds itself is informative about the determinants of mobility.

[3] The use of a linear model can be adapted to predict any continuous or discrete outcome upon some non-linear transformation, such as Tobit, logit, and probit.

Although Dang et al. (2014) and Cruces et al. (2015) validate this approach comparing it to a series of true panel estimates, Hérault and Jenkins (2019) present results from panel surveys where it does not work so well. They argue that estimates are very sensitive to the choice of cohort attributes, criticizing the use household head characteristics. They explain that household head characteristics are highly susceptible to mobility events, even when their distribution of characteristics looks similar over time.

**Remark 1**: Only individual features can be consistently linked over time. Household characteristics, such as size, number of children, and gender of the head, may change for the same individual.

Furthermore, regardless of the choice of attributes, synthetic methods contain no information to point estimate within-cohort mobility. Again, this estimation would rely on the condition that within-cohort mobility is zero.

**Remark 2**: There is no component in synthetic panel methods that point identifies within-cohort mobility when it is not zero. The larger the within-cohort component of mobility, the greater the difference between the lower bound and the true rate and the greater the bias in point estimates.

### 4.4. Difference-in-Differences

The last type of pseudo-panel analysis discussed here is the DiD. This model is often applied for the evaluation of public policies (Card & Krueger, 1994). Under the assumptions that a policy or aggregate shock is exogenous to individuals' behaviour and the difference between cohorts remains constant in the absence of this shock, one may apply it to infer causal effects. Despite this ambitious purpose, we can also use DiD models to investigate the **sensitivity of different groups to aggregate shocks**.

The DiD model can be described as follows:

$$y_{igt} = \gamma' x_g z_t + \alpha' x_g + \beta_t + u_{igt}, \tag{5}$$

where $z_t$ is an aggregate shock common to all individuals, $x_g$ is a vector of cohort attributes, $\beta_t$ represents period-specific effects, and $u_{igt}$ is the error term. In this specification, $\beta_t$ controls for the common effect of $z_t$ across cohorts, as well as other macro events, and $\alpha' x_g$ controls for underlying differences across cohorts. Hence, coefficient $\gamma$ captures whether the difference between cohorts changes along with the aggregate shock.

The necessary assumption to estimate equation (5) is that the relationship between $y_{igh}$ and $x_g$ does not change before the shock, $z_t$, takes place. This assumption can be tested by including lead interactions in the model — i.e., $x_g z_{t+1}$, $x_g z_{t+2}$, and so on.

## 5. How to Choose Cohort Attributes and Cohort Size

### 5.1. Methods

As discussed above, only fixed individual attributes should be used to construct cohorts. Still, we must examine how to work with these attributes to obtain accurate cohort means.

A **conventional approach** is to combinate attributes into cells and **calculate the cohort mean** for each cell and period, $\bar{y}_{gt}$. For example, one may split the sample into five-year intervals for the year of birth (e.g., 1940-1944, 1945-1949, ... ,1995-1999) and between genders. The goal is to make the individuals in these cells as homogenous as possible by reducing the variance of the outcome, $y_{igt}$. To do so, we may also include other attributes in the construction of cells, such as ethnicity, education, and region, and shorten the birthyear intervals. However, the finer the cells the smaller their sample size. As a result, the accuracy of cohort means decreases due to greater sampling errors.

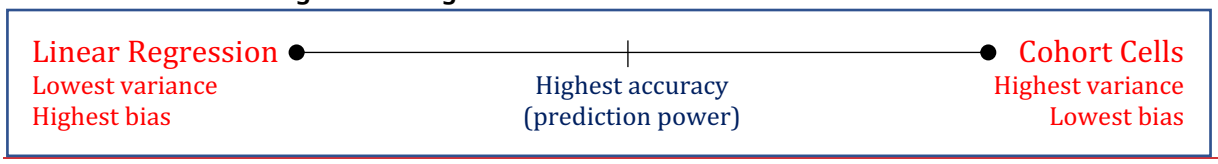
Almost all empirical studies that adopt this approach do not apply a clear statistical method to construct cohorts. Based on Verbeek and Nijman (1992, 1993), the only condition in those studies is that cohorts have at least 100 observations per period. The idea is that above 100 observations, the sampling error becomes irrelevant. However, it is not often mentioned that this magic number also depends on the variance of the

outcome, $y_{igt}$. The larger the true population variance, the bigger the required sample. Besides, when a cohort cell has less than 100 observations, those studies do not provide any rule for which attributes or intervals should be aggregated.

To overcome these issues, both Moffitt (1993) and Dang et al. (2014) adopt a more **pragmatic approach**. Instead of estimating cohort means from constructed cells, they simply estimate $y_{igt}$ using a **linear model**, where cohort attributes are independent variables. Based on goodness-of-fit statistics, such as the *F*-statistic and *R*-square, the linear regression unveils which attributes are more and less important to predict the outcome. According to Dang et al. (2014), the accuracy of the predicted outcome, $\hat{y}_{gt}$, increases with the *R*-square. Thus, the more attributes we include in the model the better, including higher-order polynomials for continuous variables such as birthyear — i.e., birthyear$^2$, birthyear$^3$, birthyear$^4$, and so on.

The problem is that the *R*-square is not a proper measure of prediction power (Hastie, et al., 2009). The *R*-square measures the goodness of fit within the sample, but it does not tell how well the estimated model performs in other random samples. For instance, consider the case with the highest possible *R*-square, which is when we include a dummy for each possible combination of attributes (i.e., one for each cohort cell). The predicted values in this regression would be equal to cohort means calculated from small cells. Albeit unbiased, this procedure yields the largest standard errors for predicted values. Following Moffitt (1993), one may get around sampling errors by simplifying the regression model. Thus, somewhere between cohort means and the simplest linear model, we must find the highest prediction power.

**Figure 1 – Range of Methods to Calculate Cohort Means**




In practice, the highest prediction power can be found using **supervised learning algorithms**, such as multivariate adaptive regression splines (MARS) and regression tree (Hastie, et al., 2009). Each of these algorithms has tune parameters that reduce their bias in exchange for higher variability and vice-versa.

From all the supervised learning algorithms, the **regression tree is particularly useful for pseudo-panel analyses** because it breaks the sample into cells creating a hierarchy of attributes.[4] That is, it gives a clear method for combining attributes in the conventional approach. Given all the possible attributes, a regression tree produces a sequence of binary splits (or nodes). At any node, a new optimal split is chosen out of every possible split on every attribute. The optimal split is the one that minimizes the variance (or the inequality) within the resulting partitions (or branches). The process of growing new branches (or splitting the sample into nodes) goes on until a node reaches a minimum size, for which new splits are not allowed, the observations in the node do not differ in any attribute, or the outcome does not vary within the node. This terminal node is also called a leaf.

Like a linear regression with too many covariates, a fully-grown tree minimizes the bias, but maximizes the prediction variance. To improve accuracy, the next step is to prune the tree by setting the tune parameters. For regression trees, there are three possible parameters to be tuned: the depth of the tree (i.e., how many branches the tree should have), the minimum number of observations per leaf (or the minimum number of observations to perform another split), and the minimum improvement at each node set by the complexity parameter (CP).[5] To pick the optimal parameters, we run the model several times to assess its accuracy.

[4] Other methods, such as random forest (RF) and K-nearest neighbour (KNN), are not as useful because their predictions do not result from a single, well-structured specification of attributes.

[5] At every node, a further split reduces the model's misclassification, but at the cost of increasing the complexity (number of splits). The CP is an arbitrary weight given to this cost. For the tree to keep growing, the reduction in misclassification must be greater than CP times the number of splits. The larger the CP, the shorter the tree.

**Remark 3**: The regression tree is a non-parametric supervised learning method that constructs cohort cells by selecting attributes and sample splits based on the accuracy of the resulting cohort means.

## 5.2. Comparison of Pseudo-Panel Mobility with True Mobility

This section presents the lower- and upper-bound mobility rates for working status, using the Continuous National Household Sample Survey in Brazil (*Pesquisa Nacional por Amostra de Domicílios Contínua*, PNADC). The advantage of using PNADC is its quarterly rotating panel. Every quarter, one-fifth of its sample is replaced by new households. As a result, for the same quarter, we can observe one-fifth of the sample in the following year. That is, one-fifth of the sample is in an annual panel, while four-fifths look like RCSs. Thus, we can compare pseudo-panel estimates in PNADC with the true mobility rate.

The pseudo-panel rates are calculated using **three different methods**. The first is the **linear model**, such as in Dang et al. (2014). The second is the **optimal regression tree**. The last is also a regression tree with no tune parameter. The only criterion that stops the tree from growing is the **minimum leaf size**. Following Verbeek and Nijman (1992, 1993), the minimum leaf size is set to be either 200 or 100. But smaller sizes are also applied: 75, 50, and 25. Still, this method is slightly different from the conventional approach of constructing cohort cells because the order of splits depends on their prediction power.

To examine mobility, the ideal method for calculating cohort means should minimize the gap between upper- and lower-bound rates, improving the explanation power of net cohort mobility rates. However, it should also guarantee that neither upper- nor lower-bound rates are misplaced. For mobility, the upper bound is misplaced when it is lower than the true mobility rate. Likewise, the upper bound is misplaced when it is lower than the true mobility rate. For immobility (i.e., the probability of staying in the same condition), the upper (lower) bound is misplaced when it is higher (lower) than the true mobility rate. Table 2 compares different methods by showing the absolute and squared differences between upper- and lower-bound rates, the mean error of the upper-bound rate as it crosses the true rate, and the mean error of the lower-bound rate as it crosses the true rate. Figure 2 and Figure 3 illustrate these assessments by showing the position of each estimated rate, given the individuals' initial condition.

Panel (a) of Table 2 indicates that the misplacement of the upper bound is very similar across methods. Indeed, **all methods yield very similar upper-bound rates**, as seen in Figure 2. The reason is that the upper-bound rate just replicates the unconditional probability of each status, and all methods produce similar probabilities on average. **The difference between methods is mostly in the lower-bound rate** (i.e., the net cohort mobility). In this regard, the lower-bound rate never crosses the true rate in the linear model because it is very close to zero (or to one for the immobility rate). However, for the same reason, this model produces the widest gap between bounds. That is, **the linear model is the least informative**.

Compared to the linear model, the optimal tree reduces the difference between bounds considerably, with almost no increase in the lower-bound error. It also slightly improves the upper-bound rate. Nevertheless, the optimal tree yields a larger difference between bounds than the unpruned tree with at least 200 observations per leaf. Compared to the unpruned trees, the optimal tree produces some small cohort cells without any increase in the lower-bound error. Still, **the lower-bound mobility in the optimal tree looks very conservative**. Perhaps, we could make it less conservative by applying another assessment score to prune the tree.

**Remark 4**: While the linear model looks uninformative, the optimal tree produces very conservative net cohort mobility rates.

**Table 2 – Performance of Methods for Working Status Mobility in Brazil, 3rd Quarter, 2012-2019**

| | Method | | | | | | |
|---|---|---|---|---|---|---|---|
| | Linear model | Optimal tree | Minimum cell size | | | | |
| | | | 200 | 100 | 75 | 50 | 25 |
| Average difference between bounds | | | | (a) **6 categories** | | | |
| Absolute | 0.26456 | 0.24445 | 0.23758 | 0.22549 | 0.21924 | 0.21064 | 0.19570 |
| Squared | 0.13709 | 0.11743 | 0.11120 | 0.10051 | 0.09525 | 0.08824 | 0.07668 |
| Upper-bound error | 0.12722 | 0.12633 | 0.12663 | 0.12670 | 0.12668 | 0.12660 | 0.12629 |
| Lower-bound error | 0.00000 | 0.00194 | 0.00548 | 0.01664 | 0.03741 | 0.06907 | 0.13840 |
| Average difference between bounds | | | | (b) **4 categories** | | | |
| Absolute | 0.35733 | 0.34122 | 0.32748 | 0.31349 | 0.30660 | 0.29695 | 0.27948 |
| Squared | 0.18046 | 0.16462 | 0.15179 | 0.13933 | 0.13337 | 0.12526 | 0.11128 |
| Upper-bound error | 0.03299 | 0.03351 | 0.03338 | 0.03326 | 0.03318 | 0.03311 | 0.03308 |
| Lower-bound error | 0.00000 | 0.00000 | 0.00000 | 0.00150 | 0.00415 | 0.00829 | 0.03263 |

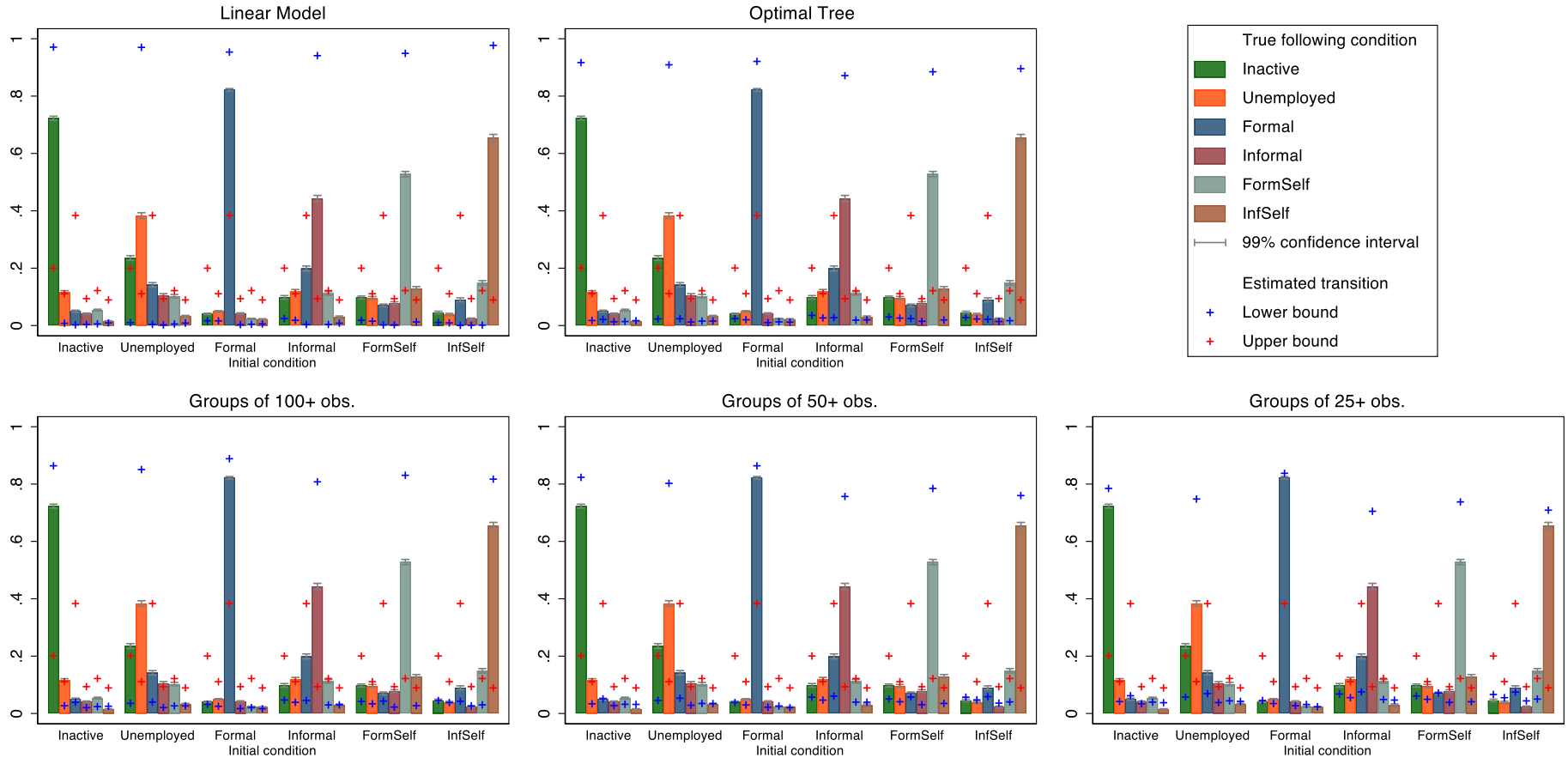
Note: Upper- and lower-bound errors give how far upper and lower bounds are misplaced when they cross to the true mobility rate. Linear model estimated using a multinomial logit. The minimum cell size method is based on an unpruned regression tree.

The smaller the minimum leaf size, the smaller the difference between bounds. However, as we reduce the minimum leaf size, the lower-bound error becomes more noticeable — i.e., the lower-bound rate starts crossing the true rate more often. In Figure 2, we observe that with a minimum leaf size of 50, the lower bound rate nearly matches many true mobility rates, being slightly higher in a few cases. With a minimum leaf size of 25, many lower-bound rates go above the true mobility rate. Therefore, considering the mobility across six categories of working status, **the minimum cohort size of 50 is as low as we can go**.

If we reduce the number of categories in working status from six to four, we observe fewer crosses in both the upper and the lower bounds. However, the average difference between bounds increases for all models. Panel (b) of Table 2 shows that even the unpruned tree with minimum leaf size of 100 yields fairly conservative rates. Figure 3 shows that a minimum size of 25 produce only a few errors, but lower-bound rates are not as close to the true rates as in the case of six categories.

Therefore, the fewer the number of categories in the outcome, the larger the difference between upper- and lower-bound rates and the smaller the minimum cohort cell size required to avoid considerable errors.

**Figure 2 – Mobility Rates for Six-Category Working Status Estimated with Different Methods Using Brazilian PNADC, 3rd Quarter, 2012-2019**




Note: The three graphs at the bottom are produced using unpruned regression trees with minimum leaf size.

# 6. Conclusion

Although pseudo-panel methods have limited interpretation, they can be extremely useful in the absence of true panel data. The most important is to understand their limitations to know when to apply and how to interpret them.

As regards labour market transitions, pseudo-panels cannot be used to assess individual risks, such as losing a job and income volatility. Individual risks are better assessed using monthly (or quarterly) surveys that have an identifiable rotating panel. For instance, Labour Force Surveys (LFS) often have this rotating panel, but they are not always easy to identify (Ribas & Soares, 2009).

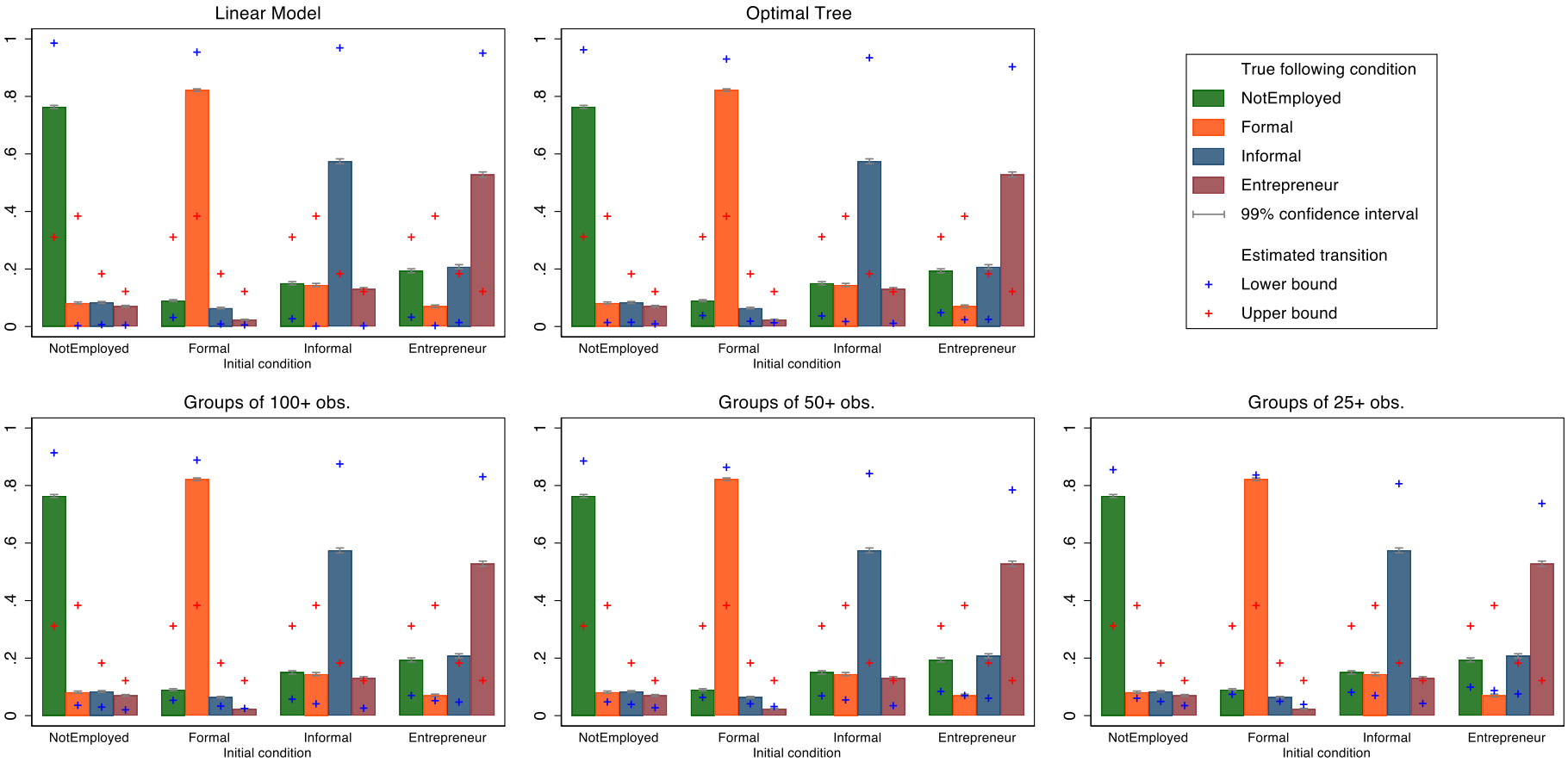
On the other hand, pseudo-panels can partially reveal mobility in the labour market. By predicting (or averaging) working conditions using cohort attributes, such as birthyear, gender, and ethnicity, we can reduce the noise created by transient events and measurement error in surveys and focus on long-term patterns. The downside is we can only observe the net cohort mobility. Namely, the overall change in cohort rates. Still, the notion of net cohort mobility is widely applied to investigate lifecycle events and long-run trends, using the APC model, and the sensitivity of demographic groups to aggregate shocks and policies, using the DiD model.

The unidentified component of mobility in a pseudo-panel is the within-cohort mobility. That is, the share of cohort members in a certain condition that are replaced by members of the same cohort who were in a different condition. Ignoring this component may lead us to draw misleading conclusions. For example, if formal and informal markets are segregated by education level, we may conclude that less educated workers never leave the informal sector when no aggregate improvement is observed.

To measure our ignorance regarding the within-cohort mobility, we can apply the synthetic panel method. This method consists of estimating bounds for individual mobility that range from the net cohort mobility (the lower bound) and the assumedly random mobility (the upper bound). The difference between bounds can be naturally large depending on how much of the individual mobility is truly explained by the net cohort mobility. Even if we apply the most accurate method to calculate net cohort mobility, its capacity to explain individual mobility is still limited by the number of attributes that can be included in the model. Only individual features that never vary over time can be used to define cohorts.

Given the limited number of cohort attributes, one may still underestimate the net cohort mobility if the model is too simple — e.g., a linear regression. One may also overestimate the net cohort mobility if the method for cohort construction is too granular, producing small cohort cells. The most accurate method is somewhere in between these two approaches. In that regard, this paper recommends the use of regression trees for the construction of cohort cells, varying the minimum cell size based on the number of categories being investigated. In the presented exercise, the minimum cell size in the regression tree can be as low as 50 if the working condition has six categories and as low as 25 if it has four. These numbers are much lower than the 100-200 proposed by Verbeek and Nijman (1992, 1993) in the conventional approach.

**Figure 3 – Mobility Rates for Four-Category Working Status Estimated with Different Methods Using Brazilian PNADC, 3rd Quarter, 2012-2019**




Note: The three graphs at the bottom are produced using unpruned regression trees with minimum leaf size.

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